many cells, some can specialize as sensors to detect prey, others as nerves to pass on the message, others as stinging cells to paralyse the prey, muscle cells to move tentacles and catch the prey, secretory cells to dissolve it and yet others to absorb the juices. We must not forget that, at least in modern bodies like our own, the cells are a clone. All contain the same genes, although different genes will be turned on in the different specialist cells. Genes in each cell type are directly benefiting their own copies in the minority of cells specialized for reproduction, the cells of the immortal germ line.

So, to the third question. Why do bodies participate in a 'bottlenecked' life cycle?

To begin with, what do I mean by bottlenecked? No matter how many cells there may be in the body of an elephant, the elephant began life as a single cell, a fertilized egg. The fertilized egg is a narrow bottleneck which, during embryonic development, widens out into the trillions of cells of an adult elephant. And no matter how many cells, of no matter how many specialized types, cooperate to perform the unimaginably complicated task of running an adult elephant, the efforts of all those cells converge on the final goal of producing single cells again—sperms or eggs. The elephant not only has its beginning in a single cell, a fertilized egg. Its end, meaning its goal or end-product, is the production of single cells, fertilized eggs of the next generation. The life cycle of the broad and bulky elephant both begins and ends with a narrow bottleneck. This bottlenecking is characteristic of the life cycles of all many-celled animals and most plants. Why? What is its significance? We cannot answer this without considering what life might look like without it.

It will be helpful to imagine two hypothetical species of seaweed called bottle-wrack and splurge-weed. Splurge-weed grows as a set of straggling, amorphous branches in the sea. Every now and then branches break off and drift away. These breakages can occur anywhere in the plants, and the fragments can be large or small. As with cuttings in a garden, they are capable of growing just like the original plant. This shedding of parts is the species's method of reproducing. As you will notice, it isn't really different from its method of growing, except that the growing parts become physically detached from one another.

Bottle-wrack looks the same and grows in the same straggly way. There is one crucial difference, however. It reproduces by releasing single-celled spores which drift off in the sea and grow into new plants.